

KubeML-Aware policy layer

**Deterministic burst generator**  
same seeds  
and manifests  
 $T, R, M, G, C, P$   
+ release gate

create

**Kubernetes API**  
complete burst held in  
SchedulingGated

observe burst

**KubeML-Aware controller**  
validate burst → normalize  
rank → FastPath check  
bounded tail balance  
→ remove gates

gate removal

**default-scheduler**  
filter, score,  
reserve, bind

normal binding

**Kubelet + containerd**  
CPU training Pods  
NumPy, buffers,  
checkpoints

structured trainer events

**Metrics API**  
fresh node  
CPU sample

headroom evidence

Gate removal changes eligibility;  
it does not select or reserve a node.

controller record

**Evidence store**  
rank, release,  
start, completion